

Marco John Lewis

Curriculum Vitae

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Research Interests

Fields Quantum computing, software verification, temporal logic

A researcher interested in understanding the interplay between quantum computing and formal verification techniques; particularly in automated verification techniques and how to adapt them to ensure the correctness of quantum hardware and software.

Positions

2024-2026 **Postdoctoral Researcher**, *INRIA (Institut national de recherche en sciences et technologies du numérique)*, Quantum Computation Structures (QuaCS).

Education

2020-2024 **PhD: Automated Verification Techniques for Quantum Computers**, *Newcastle University*, Supervisors: Paolo Zuliani and Sadegh Soudjani.

2016-2020 **MEng Mathematics and Computer Science**, *University of Bristol*.

First Class Honours

Modules: Quantum Information Theory, Quantum Computing, Types and Lambda Calculus, Advanced Algorithms, Language Engineering.

Research Outputs

Publications

2026 **Some Results on Causal Modalities in General Spacetimes**, *Marco Lewis, Nesta van der Schaaf*, In Proceedings of the Sixteenth International Conference on Advances in Modal Logic (AiML 2026).

DOI: [10.4204/EPTCS.447.33](https://doi.org/10.4204/EPTCS.447.33)

Finding Photonics Circuits via δ -Weakening SMT, *Marco Lewis, Benoît Valiron*, Verification, Model Checking, and Abstract Interpretation. VMCAI 2026. Lecture Notes in Computer Science, vol 16417. Springer, Cham.

DOI: [10.1007/978-3-032-15700-3_7](https://doi.org/10.1007/978-3-032-15700-3_7)

2025 **High-level quantum algorithm programming using Silq**, *Viktorija Bezganovic, Marco Lewis, Sadegh Soudjani, Paolo Zuliani*, In Proceedings of the 34th International Symposium on High-Performance Parallel and Distributed Computing (HPDC '25).

DOI: [10.1145/3731545.3744152](https://doi.org/10.1145/3731545.3744152)

- 2024 **Automated Verification of Silq Quantum Programs using SMT Solvers**, Marco Lewis, Paolo Zuliani, Sadegh Soudjani, In 2024 IEEE International Conference on Quantum Software (QSW), Shenzhen, China, 2024, pp. 125-134.
DOI:[10.1109/QSW62656.2024.00027](https://doi.org/10.1109/QSW62656.2024.00027)
- T-Count Optimising Genetic Algorithm for State Preparation**, Andrew Wright, Marco Lewis, Sadegh Soudjani, Paolo Zuliani, In 2024 IEEE International Conference on Quantum Software (QSW), Shenzhen, China, 2024, pp. 58-68.
DOI:[10.1109/QSW62656.2024.00020](https://doi.org/10.1109/QSW62656.2024.00020)
- Formal Verification of Quantum Programs: Theory, Tools and Challenges**, Marco Lewis, Sadegh Soudjani, Paolo Zuliani, ACM Transactions on Quantum Computing 5, 1, Article 1 (March 2024), 35 pages.
DOI:[10.1145/3624483](https://doi.org/10.1145/3624483)
- 2023 **Verification of Quantum Systems Using Barrier Certificates**, Marco Lewis, Paolo Zuliani, Sadegh Soudjani., In: Jansen, N., Tribastone, M. (eds) Quantitative Evaluation of Systems. QEST 2023. Lecture Notes in Computer Science, vol 14287. Springer, Cham.
DOI:[10.1007/978-3-031-43835-6_24](https://doi.org/10.1007/978-3-031-43835-6_24)
- [Pre-prints](#)
- 2026 **pfoqcompiler: Efficient Compilation of Quantum Control in Recursive First-Order Quantum Programs**, Mário Silva, Alexandre Guernut, Marco Lewis, HAL preprint, [hal-05738864](https://hal.archives-ouvertes.fr/hal-05738864).
- 2024 **Verification of Quantum Circuits through Discrete-Time Barrier Certificates**, Marco Lewis, Sadegh Soudjani, Paolo Zuliani, arXiv preprint, [arXiv:2408.07591](https://arxiv.org/abs/2408.07591).
- 2022 **Matrix Representation of Arbitrarily Controlled Quantum Gates**, Marco Lewis, Sadegh Soudjani, Paolo Zuliani, arXiv preprint, [arXiv:2205.02525](https://arxiv.org/abs/2205.02525).
- [Software Artifacts](#)
- 2025 **Finding Photonics Circuits via δ -weakening SMT**, Python.
Zenodo DOI:[10.5281/zenodo.17591132](https://doi.org/10.5281/zenodo.17591132)
- 2023 **SilVer: Silq Verification**, Python, D language.
Zenodo DOI:[10.5281/zenodo.11395797](https://doi.org/10.5281/zenodo.11395797)
- Verification of Quantum Systems Using Barrier Certificates**, Python.
GitHub repository: [marco-lewis/quantum-barrier-certificates](https://github.com/marco-lewis/quantum-barrier-certificates)

Dissertations

PhD Dissertation

Title *Automated verification techniques for quantum computers*
 Supervisor Prof. Paolo Zuliani, Prof. Sadegh Soudjani
 Link <http://theses.ncl.ac.uk/jspui/handle/10443/6554>

Description Investigated different approaches to verifying quantum computers. The core of my PhD was split between two lines of study: an investigation of using SMT solvers to verify Silq programs, and exploring the usage of barrier certificates for quantum computers.

[Masters Dissertation](#)

Title *Quantum Transport and Random Matrix Theory*

Supervisor Prof. Francesco Mezzadri

Description Gained an introduction to quantum chaos and knowledge of modelling quantum dots using random matrix theory. Investigated statistical properties in a non-ideal setting, which awarded a First-class mark (80%).

[Bachelor Dissertation](#)

Title *Learning to Securely Generate Keys*

Supervisor Dr. Miranda Mowbray

Description Studied how a neural network could be implemented in a cryptographic scheme. Gained experience in scientific writing, literature reading and time management, achieving a First-class mark (74%).

[Experience](#)

[Service](#)

2026 **QEST+FORMATS 2026**, *Reviewer, Artifact Reviewer.*

2025 **TACAS 2026**, *Artifact Reviewer.*

QEST+FORMATS 2025, *Artifact Reviewer.*

2024 **TACAS 2025**, *Artifact Reviewer.*

CAV 2024, *Artifact Reviewer.*

2023 **TACAS 2024**, *Artifact Reviewer.*

2021 **FoSSaCS 2022**, *Reviewer.*

ATVA 2021, *Reviewer.*

[Organisational](#)

2023-2024 **AMBER Talks Organiser**, *School of Computing*, Newcastle University.

2023-2024 **HyCoDeV Meeting Organiser**, *School of Computing*, Newcastle University.

2022-2023 **SaGE Faculty PGR Conference Committee Member**, *Faculty of Science, Agriculture & Engineering*, Newcastle University.

2021-2023 **PhD Course Representative, Chair of Student Staff Committee**, *School of Computing*, Newcastle University.

[Teaching and Supervising](#)

2023-2024 **Assistant Supervisor**, *School of Computing*, Newcastle University.

Supervising undergraduate students for dissertation project and follow up research (see *T-count Optimising Genetic Algorithm for State-preparation (TOGAS)* and *High-level quantum algorithm using Silq* in Research Outputs).

2021-2024 **Demonstrator (Teaching Assistant)**, *School of Computing*, Newcastle University.

- 2023/24 Fault Tolerant and Cyber-Physical Systems
- 2022/23 Major Project and Dissertation in Computer Science
 Introduction to Quantum Computing
 Group Project in Data Science
 Data Management and Exploratory Data Analysis
 Computing Foundations of Data Science
- 2021/22 Group Project in Data Science
 Big Data Analytics
 Data Management and Exploratory Data Analysis
 Computing Foundations of Data Science
- 2020/21 Group Project in Data Science
 Big Data Analytics
- 2020 **Introduction to Learning and Teaching in Higher Education**, *Newcastle University*.
 Part of the Vitae Researcher Development Framework
- [Miscellaneous](#)
- 2021-2022 **Team Captain**, *Newcastle University Karate Club*, Newcastle University.
- 2019-2020 **Society President**, *Combat Karate*, University of Bristol.
- 2018 **Software Intern (Summer)**, *Ghyston*, Bristol.
- 2007-Present **Martial Artist**.
 Okinawan Gojo-ryu Karate (18 years), Shotokan Karate (2 years), Muay Thai (2 years),
 Vovinam (1 year)

Research Skills

- Autonomous worker
- Fast learner and adaptable to new systems
- Problem solving
- Time management

Computer skills

- Primary PYTHON, L^AT_EX
- Skilled Microsoft Office
- Basic C, C#, D language, Docker, HASKELL, Linux